



▼ Welcome to TinyML

Litterature

Hardware

Software

Lesson plan fall 2025

▶ Part 1 | Introduction to TinyML & ML Basics

▶ Part 2 | Embedded Hardware - The Particle Photon2

▶ Part 3 | Data Acquisition

▶ Part 3.1 | Build a data acquisition system

▶ Part 5 | Feature engineering

▶ Part 5.1 | Implement feature calculation on your devic

Part 6 | Introduction to Machine Learning

▶ Part 7 | Introduction to EM Learn, ML on Microcontr

Part 7.1 | Implement a ML model trained on your fe:

▶ Part 8 | Introduction to Deep Learning

▶ Part 8.1 | Introduction to Convolutions

▶ Part 9 | Introduction to AiFES, Deep Learning on

Part 10 | Optimisation techniques, Quantisation

Part 11 | Optimisation techniques, Pruning

▶ Part 12 | Anomaly detection

▶ Semester/Exam Project

Unclaiming your device - HUSK

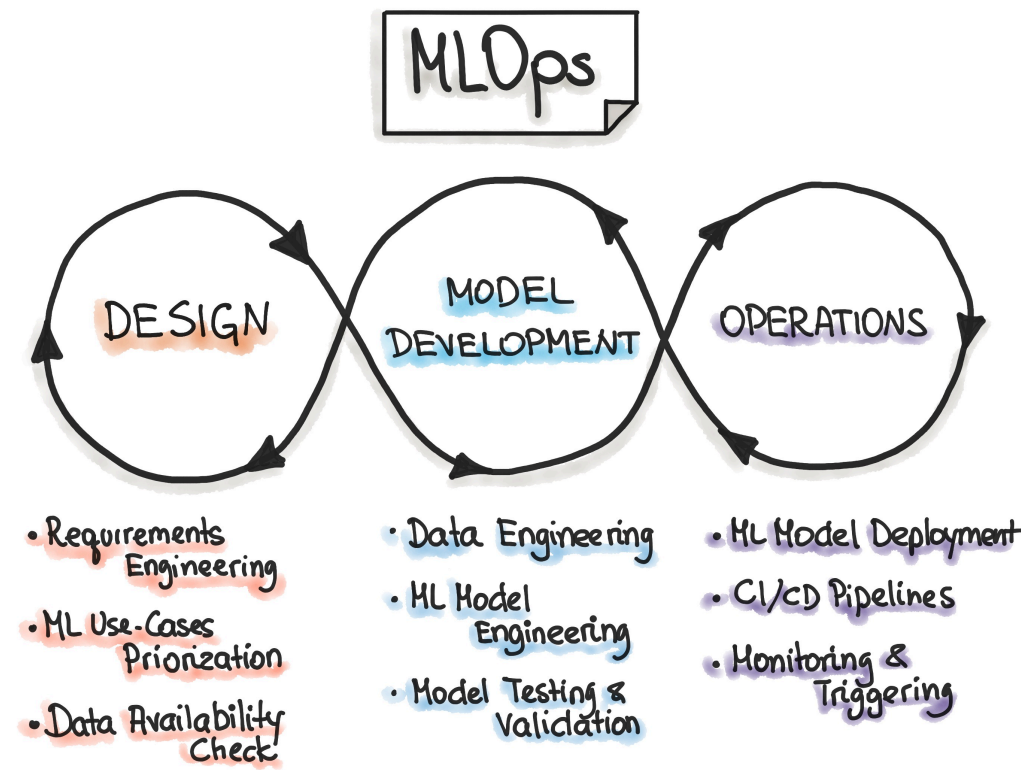
Software

TinyML Tooling:

TinyML is a fast emerging field, at the intersection of embedded systems and machine learning.

Currently there's no defined "*this is how we normally do it*" in terms of tools and processes.

TinyML practitioners are mostly inspired by the ML-OPS approach illustrated below, an iterative approach where frequent redesign, model tuning, redeployment and performance monitoring are essentials.



Course tools:

In this course we will be using the tools listed below, and likely some more...

You should install the **Course Tools** below, before we start, **Embedded ML Tools** will be installed as the course progresses

Python

Most data analysis and Machine Learning will happen using python, I recommend installing python 3.11 or newer.

Jupyter

Jupyter is an interactive environment, that allows you to run your python code in an interactive manner, in your browser, or in an IDE like VSCode. It is very well suited to experiment with data science projects and machine learning.

Google Colab

In addition to running examples using python on your local machine, we will run example code, written in python but hosted as a Google Colab. You will be able to run the examples in a modern browser such as Google Chrome, as **Google Colab notebooks**. If you can run the notebook in the link, you don't need to install anything more; if not, consider installing a recent version of Google Chrome.

Colab notebooks can also be downloaded and executed offline on your machine locally, using Jupyter.

VSCode and Particle Workbench

To program the Photon2 we need a compiler and an integrated development environment (IDE). Visual Studio Code is recommended, together with the Particle Workbench.

Embedded ML tools:

EMLearn

EMLearn is a tool, suite for converting simple trained ML models to efficient C implementations. See the appropriate learning module for details

AlfES

AlfES is a set of tools for conversion and inference of Simple Neural Network Architectures for microcontrollers. See the appropriate learning module for details.

EdgeImpulse

EdgeImpulse is an end-to-end platform for building Tiny Machine Learning applications. EdgeImpulse abstracts away all the "*tricky stuff*" and provides a binary or a library ready for download.

We may use EdgeImpulse for a few of the exercises, to speed up the process, and avoid problems with gcc, firmware download, tensorflow and python.

You should sign up / register, as well as install the CLI-tools we may need these during the course.